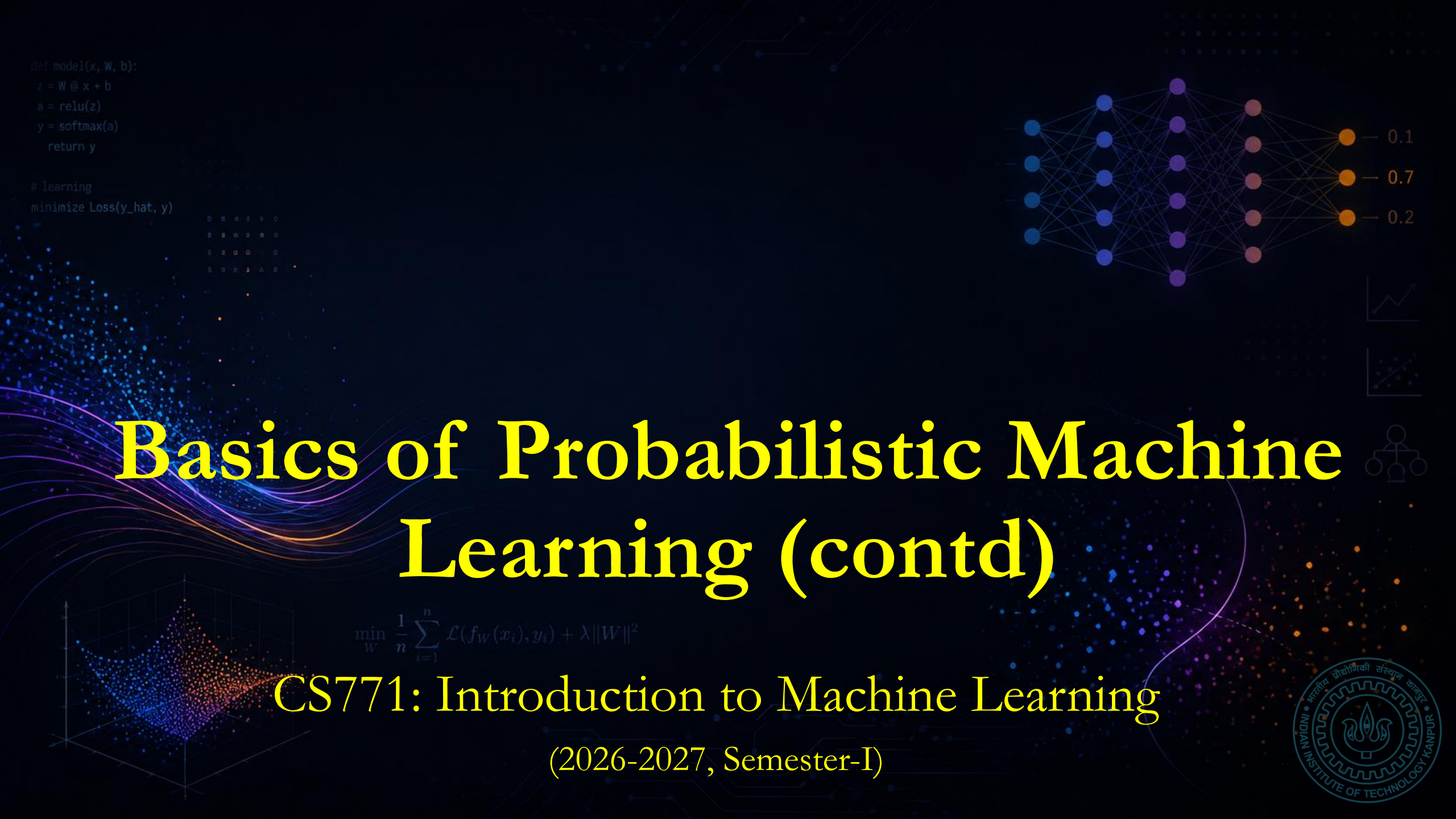


Basics of Probabilistic Machine Learning (contd)

CS771: Introduction to Machine Learning

(2026-2027, Semester-I)

Recap: Parameter Estim. in Probabilistic Models

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \log p(\mathcal{D} \mid \theta) = \arg\min_{\theta} \left[-\log p(\mathcal{D} \mid \theta)\right]$$

$$= \arg\min_{\theta} \text{NLL}(\mathcal{D}; \theta)$$

NLL is like a loss function, log-prior is like a regularizer

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \log p(\theta \mid \mathcal{D}) = \arg\max_{\theta} \left[\log p(\mathcal{D} \mid \theta) + \log p(\theta)\right]$$

$$= \arg\min_{\theta} \left[\text{NLL}(\mathcal{D}; \theta) - \log p(\theta)\right]$$

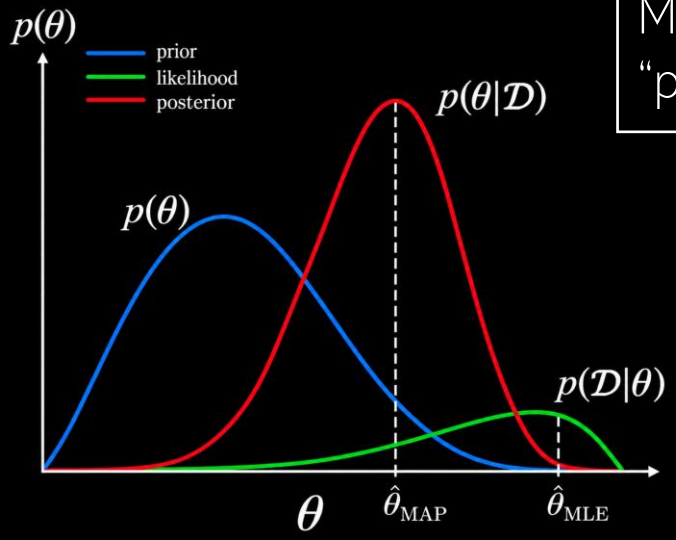
Finding the MAP doesn't require computing the full posterior; just finding its maxima

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$$

$$\propto p(\mathcal{D} \mid \theta)p(\theta)$$

Difficult to compute the full posterior exactly, except in some special cases

e.g., when likelihood and prior are conjugate to each other



MLE and MAP are only "point estimates"



The posterior provides a more holistic view of θ including the uncertainty in our estimate of θ

Recap: Bernoulli Observation Model: MLE & MAP

Observations $\mathcal{D} = \{y_1, \ldots, y_N\}$ drawn from $p(y|\theta) = \text{Bernoulli}(y|\theta)$

Assuming i.i.d. data, likelihood: $p(\mathcal{D} \mid \theta) = \prod_{n=1}^{N} p(y_n \mid \theta)$

$$\hat{\theta}_{\text{MLE}} = \arg\min_{\theta} \text{NLL} = \arg\min_{\theta} -[N_1 \log \theta + N_0 \log(1-\theta)] = \frac{N_1}{N_0 + N_1} = \frac{N_1}{N}$$

Number of 1s

Number of 0s

For MAP, we can use a Beta prior: $p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}$

Minimizing the MAP objective $[\text{NLL} - \log p(\theta)]$ gives us

$$\hat{\theta}_{\text{MAP}} = \arg\min_{\theta} -[N_1 \log \theta + N_0 \log(1-\theta) + (\alpha-1)\log\theta + (\beta-1)\log(1-\theta)]$$

$$= \frac{N_1 + \alpha - 1}{N + \alpha + \beta - 2}$$

An effect of this prior: $\alpha - 1$ and $\beta - 1$ are acting as additional number of 1s and 0s respectively (prior's hyperparameters acting as "pseudo-observations")



Bernoulli Observation Model: The Posterior

With Bernoulli likelihood $p(\mathcal{D} \mid \theta)$ and the Beta prior $p(\theta)$ the posterior distribution is also very easy to calculate in this case

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$$

$$\propto p(\theta)p(\mathcal{D} \mid \theta)$$

$$\propto \theta^{\alpha-1}(1-\theta)^{\beta-1} \times \theta^{N_1}(1-\theta)^{N_0}$$

$$= \theta^{\alpha+N_1-1}(1-\theta)^{\beta+N_0-1}$$

For this step, we can safely ignore the constant terms in prior and likelihood that don't depend on θ because $p(\theta|\mathcal{D})$ is a distribution over θ

For this easy case, even calculating the posterior up to a proportionality constant will "reveal" what the final posterior will be.

No need to worry about the marginal likelihood term $p(\mathcal{D})$ in the denominator of Bayes rule

When likelihood and prior are conjugate pair, posterior is easy to calculate and is the same distribution as the prior



Conjugate pairs are easy to identify – just notice their functional forms (e.g., see expressions of Beta and Bernoulli as a function of θ)

We can easily notice from the above that the posterior is

$$p(\theta \mid \mathcal{D}) = \text{Beta}(\theta|\alpha + N_1, \beta + N_0)$$

Thus the posterior is also Beta just like the prior, only the hyperparams got updated from $\alpha \to \alpha + N_1$ and $\beta \to \beta + N_0$

Bernoulli and Beta are a conjugate pair to each other

The Gaussian Distribution (Univariate Case)

Distribution over a scalar random variable $x \in \mathbb{R}$

PDF defined by mean $\mu \in \mathbb{R}$ and variance $\sigma^2 \in \mathbb{R}_+$ (or precision $\beta = \frac{1}{\sigma^2}$)

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \quad \text{(also written as)} \quad \mathcal{N}(x|\mu, \beta) = \sqrt{\frac{\beta}{2\pi}} \exp\left[-\frac{\beta}{2}(x-\mu)^2\right]$$

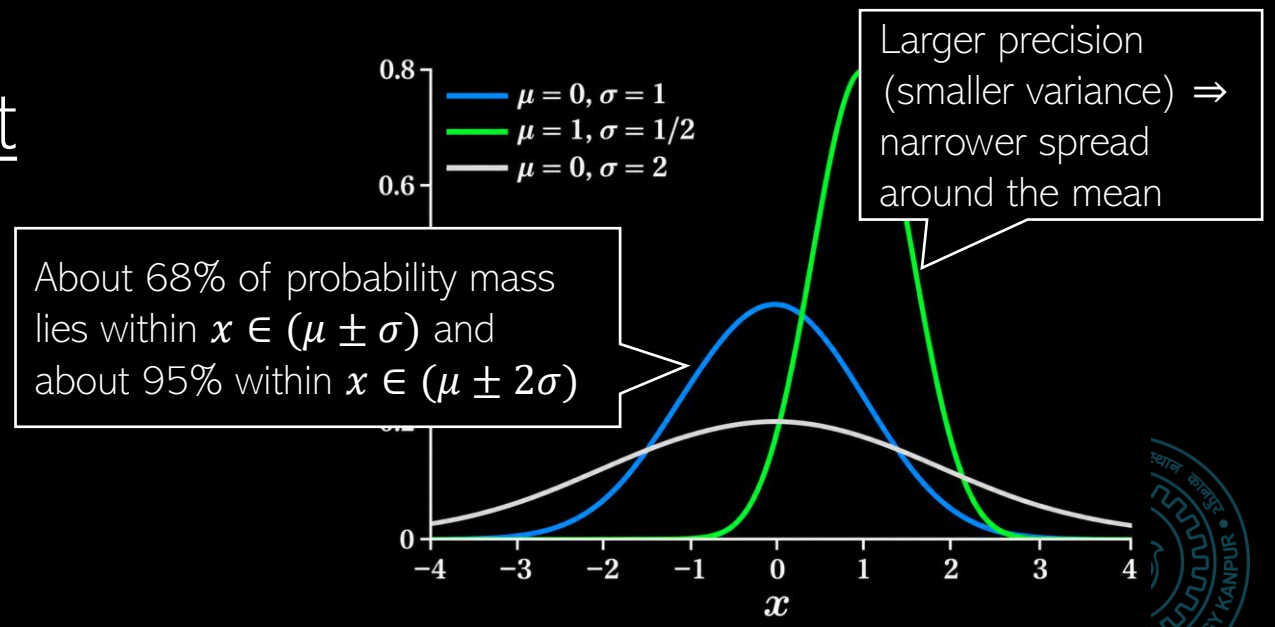
A basic but useful fact

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

$$y = \mu + \epsilon$$

$$p(y) = \mathcal{N}(y|\mu, \sigma^2)$$

A Gaussian random variable can be viewed as a "perturbation" around a fixed mean μ by adding some zero-mean Gaussian noise



About 68% of probability mass lies within $x \in (\mu \pm \sigma)$ and about 95% within $x \in (\mu \pm 2\sigma)$

Larger precision (smaller variance) ⇒ narrower spread around the mean

The Gaussian Distribution (Multivariate Case)

Distribution over a vector random variable $\boldsymbol{x} \in \mathbb{R}^D$

PDF defined by mean $\boldsymbol{\mu} \in \mathbb{R}^D$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}_+^{D \times D}$ (or its inverse: the precision matrix $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$)

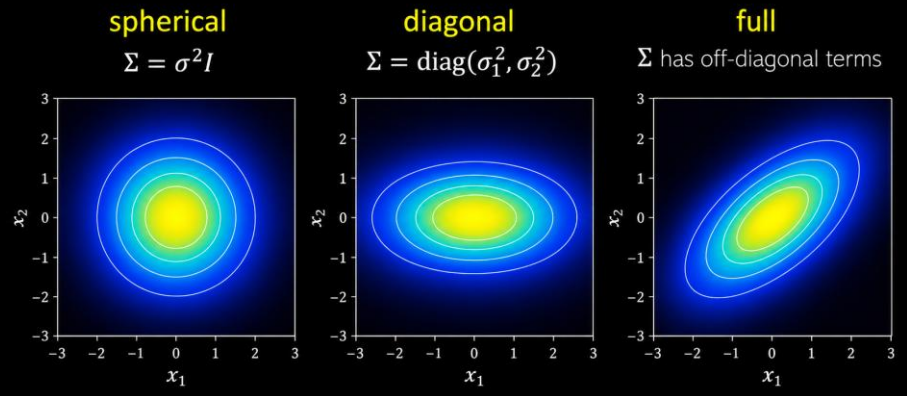
Important: Any distribution $p(x) \propto \exp\left(-\frac{1}{2}x^\top Ax + b^\top x + c\right)$ is a Gaussian $\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = \boldsymbol{A}^\top \boldsymbol{b}$ and $\boldsymbol{\Sigma} = \boldsymbol{A}^{-1}$ if A is positive-definite. Useful result to identity Gaussians when doing algebraic manipulations

$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\right]$$

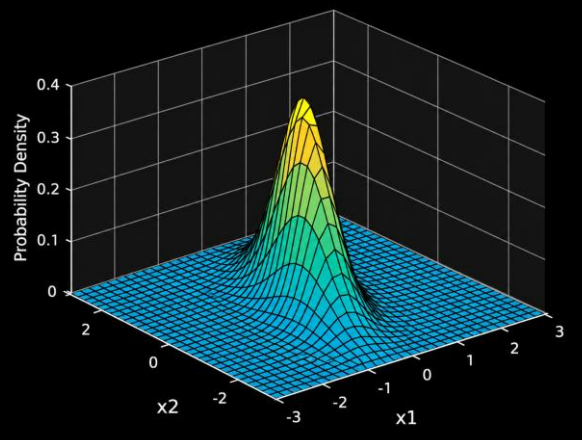
$\boldsymbol{\Sigma}$ must be positive-definite

$\boldsymbol{\Sigma}$ controls the shape of the distribution (similar to how σ^2 does it in univariate case)

Contour plots for various types of covariance matrices



2D Gaussian PDF visualized



MLE and MAP for Gaussian Observation Model

Scalar observations $\mathcal{D} = \{y_1, \ldots, y_N\}$ from $p(y|\theta) = \mathcal{N}(y|\mu, \sigma^2)$

Assuming i.i.d. observations, the NLL for the model can be written as

$$\text{NLL} = \frac{N}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{n=1}^{N}(y_n - \mu)^2$$ (exercise)

Assuming σ^2 as known, $\mu_{MLE} = \arg\min_{\mu} \text{NLL} = \frac{\sum_{n=1}^{N} y_n}{N}$ (exercise)

For MAP, we can use a Gaussian prior: $p(\mu) = \mathcal{N}(\mu|\mu_0, \sigma_0^2)$

Says that, a priori, we believe true μ to be close to μ_0

σ_0^2 is the strength of our belief

$$\mu_{MAP} = \arg\min_{\mu}[\text{NLL} - \log p(\mu)] = \frac{\frac{N}{\sigma^2}\mu_{MLE} + \frac{1}{\sigma_0^2}\mu_0}{\frac{N}{\sigma^2} + \frac{1}{\sigma_0^2}}$$ (exercise)

Exercise: Think about various scenarios: what happens to μ_{MAP} when N becomes too large/too small (also σ_0^2 and σ^2 becoming too large/too small)

Gaussian Observation Model: The Posterior

Gaussian likelihood $p(\mathcal{D} \mid \mu) = \prod_{n=1}^{N} \mathcal{N}(y_n|\mu, \sigma^2)$ and Gaussian prior $p(\mu) = \mathcal{N}(\mu|\mu_0, \sigma_0^2)$. The posterior distribution for μ will be

$$p(\mu \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \mu)p(\mu)}{p(\mathcal{D})} \propto \prod_{n=1}^{N} \exp\left[-\frac{(y_n - \mu)^2}{2\sigma^2}\right] \exp\left[-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right]$$

Not surprising that it is a Gaussian since the terms within exp are only linear and quadratic in μ

$$\propto \exp\left[-\frac{(\mu - \mu_N)^2}{2\sigma_N^2}\right] = \mathcal{N}(\mu|\mu_N, \sigma_N^2)$$

where

This posterior's mean is the same as the MAP solution (which is mode of the posterior)

Reason: Gaussian is symmetric and unimodal

$$\mu_N = \frac{\frac{\sum_{n=1}^{N} y_n}{\sigma^2} + \frac{1}{\sigma_0^2}\mu_0}{\frac{N}{\sigma^2} + \frac{1}{\sigma_0^2}}$$

and

$$\frac{1}{\sigma_N^2} = \frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}$$

This is another example of likelihood and prior being conjugate to each other



MLE for Multivariate Gaussian (Imp Result)

Given: N observations $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N\}$, each i.i.d. from a D-dim $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

MLE solutions for mean $\boldsymbol{\mu} \in \mathbb{R}^D$ and $D \times D$ covariance matrix $\boldsymbol{\Sigma}$

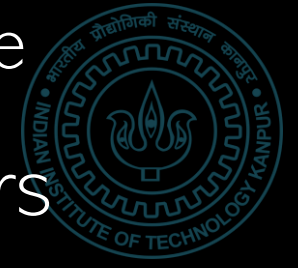
$$\boldsymbol{\mu}_{MLE} = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n$$

$$\boldsymbol{\Sigma}_{MLE} = \frac{1}{N}\sum_{n=1}^{N} (\boldsymbol{x}_n - \boldsymbol{\mu}_{MLE})(\boldsymbol{x}_n - \boldsymbol{\mu}_{MLE})^\top$$

Caveat: MLE estimation of the $D \times D$ matrix $\boldsymbol{\Sigma}$ can be unreliable when $D > N$ because it requires $O(D^2)$ parameters to be estimated

- To estimate $\boldsymbol{\Sigma}$ in such cases, we sometimes assume it to have some special form (e.g, diagonal/spherical to reduce # of parameters) or regularize

Note: MAP or posterior can also be computed using suitable priors



How the posterior (our belief) evolves..

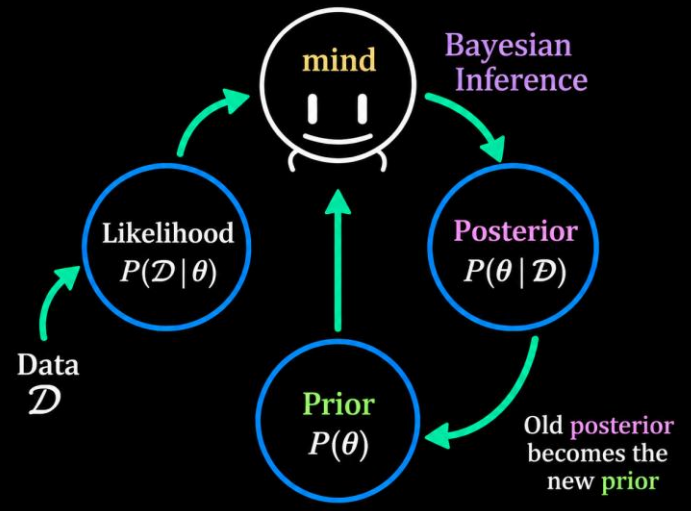
The current posterior can be updated in a naturally sequential fashion if we get an additional dataset from the same problem

$$p(\theta \mid \mathcal{D}_1) \propto p(\mathcal{D}_1 \mid \theta)p(\theta)$$

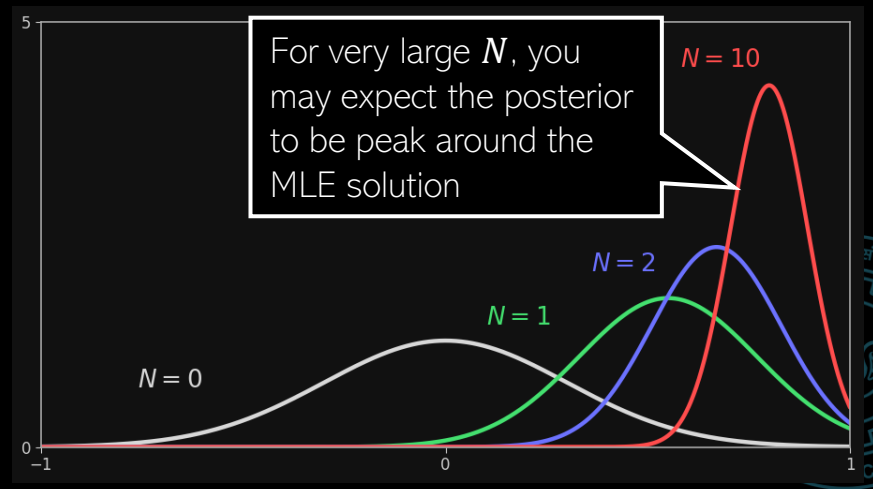
$$p(\theta, \mid \mathcal{D}_1, \mathcal{D}_2) \propto p(\mathcal{D}_2 \mid \theta)p(\theta \mid \mathcal{D}_1)$$

$$p(\theta, \mid \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3) \propto p(\mathcal{D}_3 \mid \theta)p(\theta \mid \mathcal{D}_1, \mathcal{D}_2)$$

Note how the current posterior serves as the prior for the next posterior update



Also, as we keep updating the posterior, it becomes more and more concentrated (makes sense because more evidence from data reduces uncertainty about θ)



Prediction in Probabilistic Models

In the probabilistic setup, prediction about a new y_* requires computing the predictive distribution which in general refers to this distribution

$$p(y_* \mid \mathcal{D})$$

This distribution is conditioned on training data $\mathcal{D}$ (and any other inputs/features, hyperparams, etc that y_* depends on)

If we have a posterior $p(\theta \mid \mathcal{D})$, the predictive distribution will be

$$p(y_* \mid \mathcal{D}) = \int p(y_* \mid \theta)\, p(\theta \mid \mathcal{D})\, d\theta$$

Except for some simple conjugate models, in general, this integral may need approximation

Integral does the weighted averaging over all possible values of θ

Predict using a particular value of θ

Weigh the prediction made using this value of θ by the posterior density of θ taking this value

If we only have a point estimate (MLE/MAP) $\hat{\theta}$

$$p(y_* \mid \mathcal{D}) \approx p(y_* \mid \hat{\theta})$$

A somewhat crude approximation since we are ignoring the uncertainty about θ and using a single ("point) estimate $\hat{\theta}$

But easy to compute and sometimes, it may be good enough for our needs



Fully Bayesian Prediction vs Ensemble Approxn.

Recall the general form of the predictive distribution

Also called the fully Bayesian prediction and the quantity as "posterior predictive distribution" (PPD)

$$p(y_* \mid \mathcal{D}) = \int p(y_* \mid \theta) p(\theta \mid \mathcal{D}) \, d\theta$$

Possible in the Bernoulli and Gaussian examples we saw earlier (try as an exercise)

In some simple cases, the above can be computed analytically

If the integral is intractable, we may approximate it as follows

Draw M random samples i.i.d. from the posterior

$$\theta^{(1)}, \theta^{(2)}, \ldots, \theta^{(M)} \overset{\text{iid}}{\sim} p(\theta \mid \mathcal{D})$$

Compute prediction $p(y_*|\theta)$ using each of the M samples and compute the average

$$p(y_* \mid \mathcal{D}) \approx \frac{1}{M} \sum_{m=1}^{M} p\left(y_* \mid \theta^{(m)}\right)$$

The ensemble nature of doing the predictions is another justification of why Bayesian predictions are more robust than the predictions using MLE (and even MAP, which is less prone to overfitting than MLE because of the presence of the prior which acts as a regularizer, but is still a point estimate)